
Pursuing a **Minor in AI & Data Science** offered by the Centre for Machine Intelligence & Data Science, IIT Bombay

SCHOLASTIC ACHIEVEMENTS

- Achieved **SPI of 10/10** in 4th semester | Secured **AA/AB** grades in **23 courses** till date for exceptional performance
- Selected as a **Teaching Assistant** for **120+** students for Electro-Chemistry Course based on proficiency in the subject

PROFESSIONAL EXPERIENCE

Walmart Global Tech India | *Software Engineering Intern*

(May' 25 - Jul' 25)

*Received **Pre-Placement Offer** and **Letter of Recommendation** for overall performance & contribution to the firm*

- Built **Spark-Scala** pipelines to transform raw data into curated tables powering **fraud detection** in Canada & Mexico
- Developed Freemarker **SQL CDDs** on **BigQuery**, integrating ML risk scores into product dashboards & analyst tools
- Worked on **5+ features**, including Linked Parameter & Risky Driver, contributing to **\$10M+** annual fraud **savings** in CA
- Reduced **12+ man-hours** per cycle by **automating** cross-environment data transfers using Python for business operations

KEY PROJECTS

AI-Powered Retrieval-Augmented Chatbot | *Prof Abir De* | *CS 419*

(Aug '24 - Oct '24)

*Developed an **AI chatbot** using hybrid retrieval and offline language models for accurate access to private information*

- Built AI-powered **RAG chatbot** using **TF-IDF/SVM** and semantic search, achieving **83%** classification accuracy
- Integrated offline **LLM** via **Ollama** with applied prompt engineering, reducing irrelevant response by approximately **20%**
- Engineered hybrid retrieval pipeline using cosine similarity, acronym expansion, and chunking, boosting precision by **15%**

AI Agent for Internal Ops Workflow Automation | *Analytics Club* | *IIT Bombay*

(Jul '24 - Aug '24)

*Built an **LLM-based agent** that routes user queries to correct internal tool (docs, queries, or summaries) based on intent*

- Implemented a basic AI agent loop that selects the right action (lookup, query, or summarization) before responding
- Integrated a local LLM via Ollama with simple tool interfaces to enable offline, privacy-preserving execution of workflows
- Automated a multi-step internal workflow by combining retrieval, reasoning, and action execution in a single agent flow

Credit Risk Modelling using ML | *Prof Biplab* | *GNR 638, IIT Bombay*

(Dec '24 - Jan '25)

*Predicted **credit approval status** using demographic, financial, and credit features*

- Performed comprehensive **EDA** including **Chi-square test** for categorical variables, **sequential VIF** for multicollinearity in numerical columns, and **ANOVA test** for numerical variables to optimize **feature selection** for credit risk modeling
- Implemented **Random Forest**, **Decision Trees**, & **XGBoost** for analysis, XGBoost achieving the highest performance
- **Fine-tuned** hyperparameters via **GridSearchCV**, increasing accuracy from **85%** to **91%** with an **F1-score** of **0.901**

Click Through Rate Prediction for Card Offers | *American Express*

(Jun '25 - July '25)

*Achieved **Rank 1** in **IIT Bombay campus round** among participating teams in **CTR prediction** for credit card offers*

- Performed feature engineering to shortlist **360+** features into user and offer-specific categories; created additional features
- Designed and trained a **Two-Tower network** on user and offer features improving **mAP@7** score from **0.43** to **0.51**
- Evaluated multiple models, including **XGBoost** on Amex CTR data to benchmark their **mAP@7** against Tower model

POSITIONS OF RESPONSIBILITY

Institute Internship Coordinator | *Placement Office, IIT Bombay*

(Jun '24 - May '25)

*Entrusted with streamlining internship procedure for **2500+** students and **270+** recruiters as a part of **36-member team***

- Achieved **38% YoY growth** in Day-1 internship selections-the **highest ever**-through trend analysis & targeted outreach
- Spearheaded a team of **10+** interview coordinators to handle on-ground operations of the institute placement interviews
- Liaised with **270+** firms; expanded non-core database by **150+** new firms and conducted **12+** aptitude and coding tests

TECHNICAL SKILLS & COURSES UNDERTAKEN

Programming Languages	C, C++, Python, Java, R, MATLAB, SQL, $\LaTeX$,
Python Libraries	NumPy, Pandas, SkLearn, PyTorch, Matplotlib, Seaborn
Relevant Courses	Artificial Intelligence & Data Science, Deep Learning for Computer Vision

EXTRACURRICULARS

- Bagged **Silver Medal** in **Badminton**, Inter-Hostel GC IITB | **Ranked 2nd** in the **Kabaddi**, Inter-Hostel GC IITB
- Volunteered with **NSS** to teach JEE Aspirants | Cleared **2.5 tonnes** of waste in Versova and Powai Lake clean-up drives
- Mentored **10+** students in **2 teams** for AI/ML projects with the assistance of **WnCC** and **MnP club**, **IIT Bombay**